OpenADMET CYP Inhibition Blind Challenge

Model Report

Direct inhibition prediction (regression) - Team gogocyptron - public data only

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Overview

Our approach was to develop an ensemble consisting of 3 types of models with the aim of capturing complementary information. In parallel with the challenge, we developed Atom-JEPA-Bio and subsequently added it to the ensemble with a fixed 10% contribution for the final submission.

Models

Our initial ensemble combined:

- **Pretrained 3D molecular models:** Atom-JEPA, AllScAIP and PET, fine-tuned for this challenge.
- **A 2D molecular graph model:** Chemprop.
- **A descriptor-based model:** LightGBM using concatenated Morgan fingerprints and RDKit descriptors.

Atom-JEPA-Bio builds on Atom-JEPA through a second pretraining stage that combines its original self-supervised objectives with supervision from approximately 1.5 million bioassay molecules. We describe this approach in our Atom-JEPA-Bio blog post.

Data

The challenge regression dataset contained **4,905 molecules and 6,525 observed pIC50 measurements** across CYP1A2, CYP2C9, CYP2D6 and CYP3A4. This was the complete CYP training dataset for AllScAIP, PET and LightGBM.

For Chemprop and original Atom-JEPA, we supplemented these measurements with three groups of auxiliary prediction tasks:

- **Challenge single-concentration measurements:** four $\log_2$ fold-change endpoints, one for each challenge enzyme.
- **ChEMBL CYP inhibition data:** five pIC50 endpoints covering the four challenge enzymes and CYP2C19.

- **Veith screening data:** five maximum-response endpoints from the PubChem CYP inhibition screen, AID 1851, covering the same five enzymes.

These data were assembled using the public datasets distributed through SuperCowPowers' Workbench. We treated the auxiliary measurements as separate prediction tasks rather than merging them with the challenge pIC50 labels. Together, they provided **18 tasks across 31,669 molecules**. Atom-JEPA used 31,523 of these molecules after excluding 146 auxiliary molecules without usable conformers; all challenge molecules were retained.

Missing labels were masked during training, and auxiliary molecules matching held-out or blinded-test molecular parents were excluded.

Atom-JEPA-Bio's bioassay pretraining data were separate from these CYP training datasets.

Training and ensemble construction

During model development, we used five-fold cross-validation and separate inner validation splits to tune hyperparameters using MA-ST-RAE. We fine-tuned the pretrained 3D models and averaged their predictions across multiple conformers. After selecting the training settings, we trained the final models on their respective full training datasets, without reserving a validation set.

To combine the original five models, we optimized separate ensemble weights for each enzyme using predictions on 981 held-out molecules. Following the stacking principle described by Breiman (1996), we required non-negative weights that sum to one. We used linear programming to minimize the soft-threshold absolute-error objective.

Subsequently for the final submission, we then assigned Atom-JEPA-Bio a fixed 10% weight for every enzyme and scaled the optimized ensemble to 90%. Finally, we applied endpoint-specific calibration to the test predictions using the approach from SuperCowPowers' Workbench.